

Artificial Neural Network

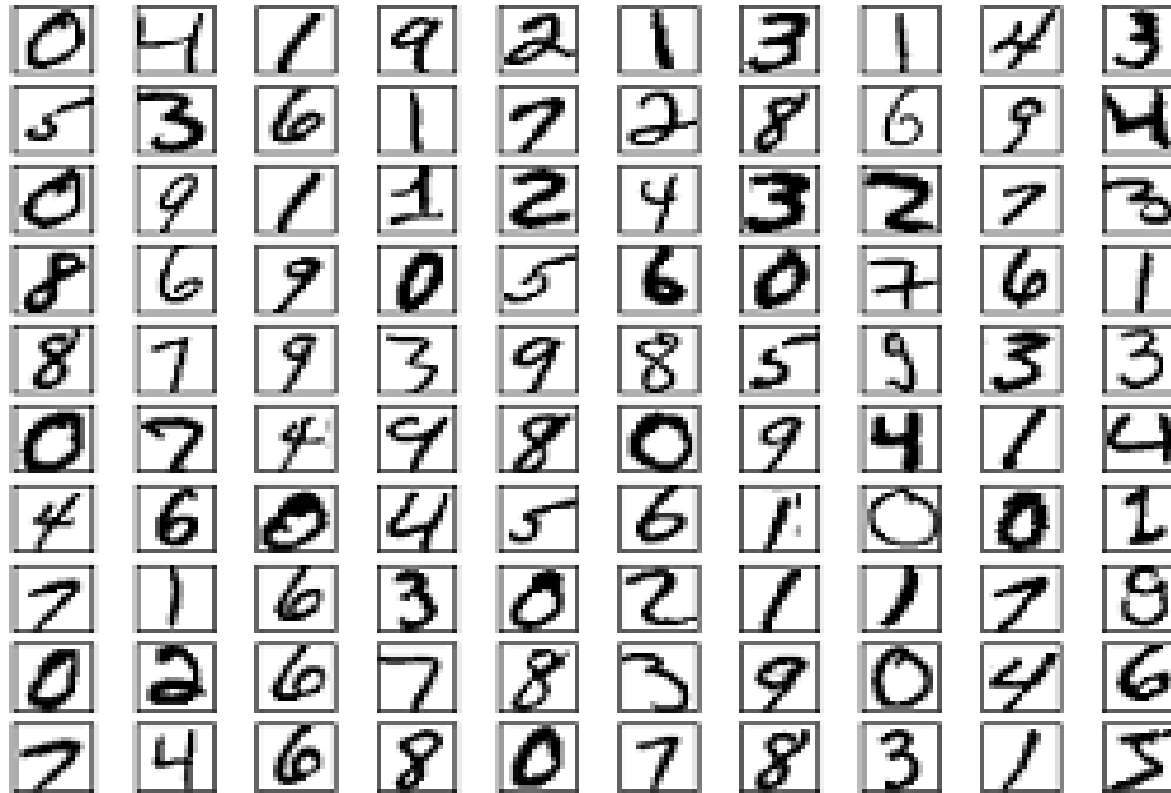
Neural Network

- Consider the following sequence of handwritten digits:

504192

- Most people effortlessly recognize those digits as 504192.
- What happens if you attempt to write a program to recognize digits like those above????

Neural Network



Training Examples

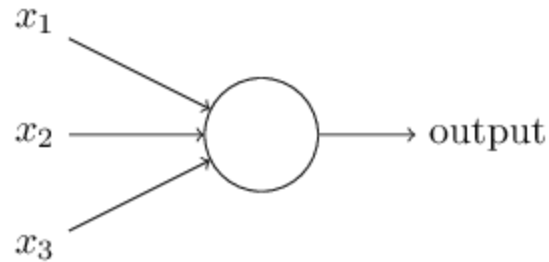
But this short program can recognize digits with an accuracy over 96 percent, without human intervention.

Neural Network

- A neural network is a model of reasoning based on the human brain.
- The human brain consists of interconnected set of nerve cells or basic information processing unit called neuron.

Perceptron

- *perceptron* – A simple ANN which consists of a single neuron.



- A perceptron takes several binary inputs, $x_1, x_2, \dots$, and produces a single binary output:

Perceptron

- *weights*, $w_1, w_2, \dots$, real numbers expressing the importance of the respective inputs to the output.
- The neuron's output, 0 or 1, is determined by whether the weighted sum $\sum_j w_j x_j$ is less than or greater than some *threshold value*.
- Just like the weights, the threshold is a real number which is a parameter of the neuron. To put it in more precise algebraic terms:

$$\text{output} = \begin{cases} 0 & \text{if } \sum_j w_j x_j \leq \text{threshold} \\ 1 & \text{if } \sum_j w_j x_j > \text{threshold} \end{cases} \quad (1)$$

- A way you can think about the perceptron is that it's a device that makes decisions by weighing up evidence.

Perceptron

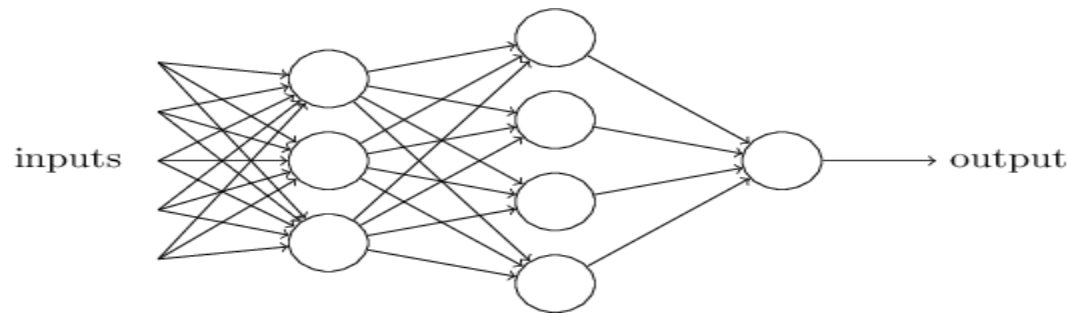
- Suppose the weekend is coming up, and you've heard that there's going to be a pizza festival in your city. You like pizza, and are trying to decide whether or not to go to the festival. You might make your decision by weighing up three factors:
- Is the weather good?
- Do your friends want to go too?
- Is the festival near public transit? (You don't own a car).
- We can represent these three factors by corresponding binary variables x_1, x_2 , and x_3 .
- $x_1=1$ if the weather is good, and $x_1=0$ if the weather is bad.
- Similarly, $x_2=1$ if your friends want to go, and $x_2=0$ if not.
- And similarly again for x_3 (public transit).

Perceptron

- Say, $w_1=6$ for the weather, and $w_2=2$ and $w_3=2$ for the other conditions.
- The larger value of w_1 indicates that the weather matters a lot to you, much more than whether your friends join you, or the nearness of public transit.
- Finally, suppose you choose a threshold of 5 for the perceptron.
- With these choices, the perceptron implements the desired decision-making model.
- It makes no difference to the output whether your friends want to go, or whether public transit is nearby.
- By varying the weights and the threshold, we can get different models of decision-making.

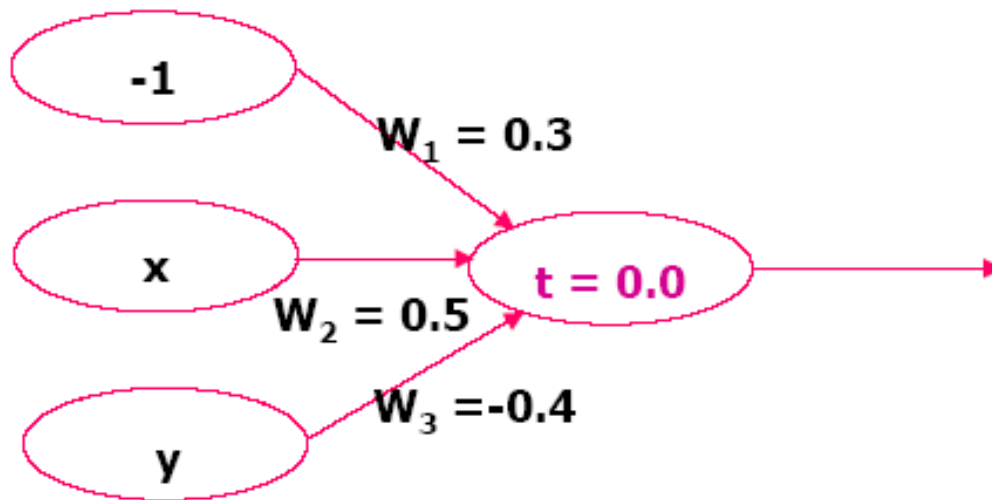
Multilayer Perceptrons

- In this network, the first column of perceptrons - what we'll call the first *layer* of perceptrons - is making three very simple decisions, by weighing the input evidence.



- Each of the perceptrons in 2nd layer is making a decision by weighing up the results from the first layer of decision-making.
- In this way a perceptron in the second layer can make a decision at a more complex and more abstract level than perceptrons in the first layer.
- And even more complex decisions can be made by the perceptron in the third layer.

How perceptron learn?



For AND		
A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

I_1	I_2	I_3	Summation	Output
-1	0	0	$(-1 \cdot 0.3) + (0 \cdot 0.5) + (0 \cdot -0.4) = -0.3$	0
-1	0	1	$(-1 \cdot 0.3) + (0 \cdot 0.5) + (1 \cdot -0.4) = -0.7$	0
-1	1	0	$(-1 \cdot 0.3) + (1 \cdot 0.5) + (0 \cdot -0.4) = 0.2$	1
-1	1	1	$(-1 \cdot 0.3) + (1 \cdot 0.5) + (1 \cdot -0.4) = -0.2$	0

How perceptron learn?

- Make small adjustment in the wights to reduce the difference between actual and desired output.
- The initial weights are randomly assigned in between $[-0.5, 0.5]$

Notation for ANN

Target value: When training the net is required to provide with the desired output as well as input. In case of AND function the target value will be 1 for [1,1]

Output (y): The output value from the neuron

I_i : Inputs being presented to the neuron

w_j : Weight from the input neuron to the output neuron

LR: The learning rate. This dictates how quickly the network converge. It is set by a matter of experimentation. It is typically 0.1.

- Error = Target – Output
- If Error < > 0 then
- $\Delta W_j = \text{new } W_j - \text{old } W_j = \text{LR} * I_j * \text{Error}$
- $\text{new } W_j = \text{old } W_j + \Delta W_j$

Learning Algorithm Example

- To Learn AND.
- Given that initial weight 0,0.4, threshold =0.3, Learning rate=0.25,

w_1	w_2	θ	x_1	x_2	$\sum w_i x_i$	y	t	Δw_1	Δw_2
0.0	0.4	0.3	0	0	0	0	0	0	0